GATE ME 90 Days Challenge

Day 89/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
All Subjects	**Final Formula Revision**	**8 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on a final one-stop formula sheet across the whole syllabus — every solution opens with the formula it uses: permutations with repeated letters, eigenvalue sum and product, Newton–Raphson, restitution, principal stresses, Euler's effective lengths, piston acceleration, transmissibility, fillet-weld throat, pump power and affinity laws, lumped cooling, ideal-gas entropy change, Merchant's shear angle and the M/M/1 queue. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 NAT

The number of distinct arrangements of all the letters of the word ENGINEER is _____

Q2 MCQ

A 3 × 3 matrix A has trace 10, and two of its eigenvalues are 2 and 3. The determinant of A is

(A) 6 **(B)** 30
(C) 25 **(D)** 60

Q3 NAT

A root of $x^3 - 2x - 5 = 0$ is found by the Newton–Raphson method starting from $x_0 = 2$. The first iterate x_1 (round off to 2 decimal places) is _____

Q4 MCQ

A ball is dropped from a height of 8 m onto a hard floor. The coefficient of restitution between the ball and the floor is 0.5. The height reached by the ball after its second bounce is

(A) 2 m **(B)** 1 m
(C) 4 m **(D)** 0.5 m

Q5 MCQ

At a point in plane stress, $\sigma_x = 50$ MPa (tension), $\sigma_y = 10$ MPa (compression) and $\tau_{xy} = 40$ MPa. The principal stresses are

(A) 70 MPa and −30 MPa **(B)** 60 MPa and −20 MPa
(C) 50 MPa and −10 MPa **(D)** 90 MPa and −50 MPa

Q6 NAT

A solid circular steel strut of 40 mm diameter and 2.5 m length is fixed at one end and pinned at the other. Taking E = 200 GPa, its Euler buckling load (in kN) is _____ (round off to 1 decimal place)

Q7 MCQ

In a slider–crank mechanism the crank is 50 mm long, the connecting rod is 200 mm long and the crank turns uniformly at 100 rad/s. The acceleration of the piston when the crank is at inner dead centre (θ = 0°) is

(A) 500 m/s²
(B) 375 m/s²
(C) 625 m/s²
(D) 531 m/s²

Q8 MCQ

An undamped machine mounting runs at three times its natural frequency. The force transmissibility is

(A) 0.333
(B) 0.125
(C) 0.111
(D) 1.125

Q9 NAT

Two plates are joined by fillet welds of 8 mm leg size with a total weld length of 150 mm. The permissible shear stress in the weld is 90 MPa. The safe load the joint can carry (in kN) is _____ (round off to 2 decimal places)

Q10 NAT

A pump delivers 0.05 m³/s of water against a total head of 40 m. Its overall efficiency is 75 %. Taking g = 9.81 m/s², the power required to drive the pump (in kW) is _____ (round off to 2 decimal places)

Q11 MCQ

The speed of a centrifugal pump is doubled (same impeller, same fluid, similar operating point). Its discharge, head and power become, respectively,

(A) 2, 4 and 8 times
(B) 2, 2 and 4 times
(C) 4, 2 and 8 times
(D) 2, 4 and 16 times

Q12 NAT

A small metal sphere at 200 °C is suddenly placed in a fluid at 20 °C. Lumped analysis applies. Its temperature (in °C) after a time equal to two time constants is _____ (round off to 2 decimal places)

Q13 MCQ

Air (c_p = 1.005 kJ/kg·K, R = 0.287 kJ/kg·K) goes from 1 bar and 300 K to 3 bar and 500 K. Its specific entropy change is about

(A) 0.513 kJ/kg·K
(B) 0.315 kJ/kg·K
(C) 0.828 kJ/kg·K
(D) 0.198 kJ/kg·K

Q14 MCQ

In orthogonal cutting with a tool of rake angle 8°, the friction angle at the tool–chip interface is 36°. The shear angle given by Merchant's theory is

(A) 31°
(B) 22°
(C) 40°
(D) 27°

Q15 NAT

Trucks arrive at a single loading bay in a Poisson stream at 15 per hour, and the loading time is exponential with a mean rate of 20 per hour (M/M/1). The average time a truck waits in the queue before loading starts (in minutes) is _____

Answer Key

Q	Type	Answer
Q1	NAT	3360
Q2	MCQ	(B) 30
Q3	NAT	2.10 (accept 2.09 to 2.11)
Q4	MCQ	(D) 0.5 m
Q5	MCQ	(A) 70 MPa and −30 MPa
Q6	NAT	79.4 (accept 78.8 to 80.0)
Q7	MCQ	(C) 625 m/s²
Q8	MCQ	(B) 0.125
Q9	NAT	76.36 (accept 75.98 to 76.74)
Q10	NAT	26.16 (accept 26.03 to 26.29)
Q11	MCQ	(A) 2, 4 and 8 times
Q12	NAT	44.36 (accept 44.14 to 44.58)
Q13	MCQ	(D) 0.198 kJ/kg·K
Q14	MCQ	(A) 31°
Q15	NAT	9

Detailed Solutions

Q1 **Answer: 3360**

Formula: Arrangements of n objects with p, q, … identical = n!/(p! q! …).

Concept: ENGINEER has 8 letters: E three times, N twice, G, I and R once each.

Arrangements = 8!/(3! × 2!) = 40 320/12 = **3360**.

Exam tip: Related formulas: circular arrangements of n distinct objects (n − 1)!; selections ${}^{n}C_{r}$ = n!/[r!(n − r)!]; arrangements ${}^{n}P_{r}$ = n!/(n − r)!.

Q2 **Answer: (B) 30**

Formula: $\Sigma\lambda_i$ = trace A (sum of diagonal entries); $\Pi\lambda_i$ = det A.

Concept: The third eigenvalue follows from the trace, and then the determinant is the product of all three.

$\lambda_3 = 10 - 2 - 3 = 5$, so det A = 2 × 3 × 5 = **30 (B)**.

Exam tip: Eigenvalues of A^{-1}, A^k, A + cI and kA are 1/λ, λ^k, λ + c and kλ; eigenvalues of a triangular matrix are its diagonal entries.

Q3 **Answer: 2.10 (accept 2.09 to 2.11)**

Formula: $x_{n+1} = x_n - f(x_n)/f'(x_n)$.

Concept: $f(x) = x^3 - 2x - 5$ and $f'(x) = 3x^2 - 2$.

$f(2) = 8 - 4 - 5 = -1$ and $f'(2) = 12 - 2 = 10$, so $x_1 = 2 - (-1/10) =$ **2.10** (true root 2.0946).

Exam tip: NR converges quadratically for a simple root. Bisection halves the interval each step; regula falsi uses $x = (a\,f(b) - b\,f(a))/(f(b) - f(a))$; the secant method uses the same formula with the two latest points.

Q4 **Answer: (D) 0.5 m**

Formula: e = (velocity of separation)/(velocity of approach); for a ball dropped from h, the rebound height after n bounces is $h_n = e^{2n}h$.

Concept: Each impact multiplies the speed by e, and the height ($\propto v^2$) by e^2.

$h_2 = 0.5^4 \times 8 = 8/16 =$ **0.5 m (D)**.

Exam tip: After the first bounce the height is $e^2h = 2$ m. Linear momentum is conserved in a collision of two free bodies; kinetic energy is conserved only if e = 1.

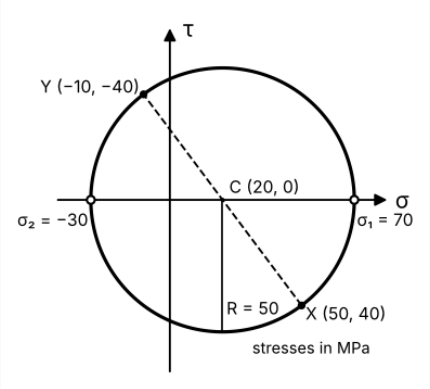
Q5 **Answer: (A) 70 MPa and -30 MPa**

Formula: $\sigma_{1,2} = (\sigma_x + \sigma_y)/2 \pm \sqrt{\{[(\sigma_x - \sigma_y)/2]^2 + \tau_{xy}{}^2\}}$; $\tan 2\theta_p = 2\tau_{xy}/(\sigma_x - \sigma_y)$.

Concept: The centre of Mohr's circle is at the mean normal stress and its radius equals the maximum in-plane shear stress.

Centre = $(50 - 10)/2 = 20$ MPa; radius = $\sqrt{(30^2 + 40^2)} = 50$ MPa. $\sigma_1 = 20 + 50 = 70$ MPa, $\sigma_2 = 20 - 50 = -30$ MPa: **(A)**.

Exam tip: Check: $\sigma_1 + \sigma_2 = \sigma_x + \sigma_y = 40$ MPa (first stress invariant). Maximum in-plane shear = radius = 50 MPa.



Q6 **Answer: 79.4 (accept 78.8 to 80.0)**

Formula: $P_{cr} = \pi^2 EI/L_e^2$; $L_e = L$ (both ends pinned), $2L$ (fixed–free), $L/2$ (both fixed), $L/\sqrt{2}$ (fixed–pinned).

Concept: For a solid circle $I = \pi d^4/64 = \pi \times 40^4/64 = 1.257 \times 10^5$ mm⁴.

With $L_e = L/\sqrt{2}$: $P_{cr} = 2\pi^2 EI/L^2 = 2\pi^2 \times 200\,000 \times 1.257 \times 10^5/2500^2 = 79\,380$ N = **79.4 kN**.

Exam tip: Relative to both ends pinned, the buckling loads are in the ratio 1/4 (fixed–free) : 1 : 2 (fixed–pinned) : 4 (both fixed). Slenderness ratio = L_e/k with $k = \sqrt{(I/A)}$.

Q7 **Answer: (C) 625 m/s²**

Formula: $a_p \approx r\omega^2[\cos\theta + (\cos 2\theta)/n]$, with $n = l/r$; piston velocity $v_p \approx r\omega[\sin\theta + (\sin 2\theta)/(2n)]$.

Concept: $n = 200/50 = 4$ and $r\omega^2 = 0.05 \times 100^2 = 500$ m/s².

At $\theta = 0°$: $a_p = 500(1 + 1/4) =$ **625 m/s² (C)**; at outer dead centre ($\theta = 180°$) it is $500(1 - 1/4) = 375$ m/s².

Exam tip: The inertia force of the reciprocating mass m is $F = m\,a_p$; its primary part is $mr\omega^2 \cos\theta$ and its secondary part $mr\omega^2 \cos 2\theta/n$.

Q8 **Answer: (B) 0.125**

Formula: $TR = \sqrt{[1 + (2\zeta r)^2]}/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$, with $r = \omega/\omega_n$; for $\zeta = 0$, $TR = 1/|1 - r^2|$.

Concept: Isolation ($TR < 1$) needs $r > \sqrt{2}$.

$TR = 1/|1 - 9| = 1/8 =$ **0.125 (B)**; i.e. 12.5 % of the force reaches the foundation.

Exam tip: For $r > \sqrt{2}$, adding damping increases TR. Magnification factor (amplitude) $X/X_{st} = 1/\sqrt{[(1 - r^2)^2 + (2\zeta r)^2]}$, which for $\zeta = 0$ has the same value here.

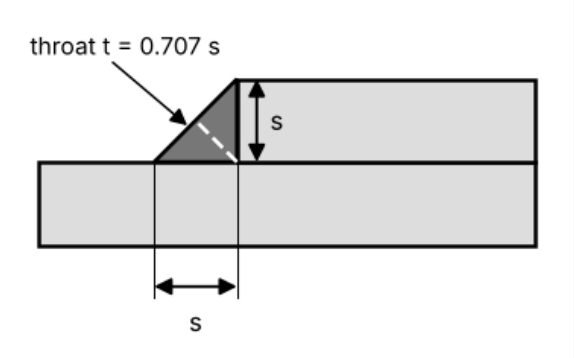
Q9 **Answer: 76.36 (accept 75.98 to 76.74)**

Formula: Fillet-weld strength $P = \tau \times t \times L$, where the throat thickness $t = s \cos 45° = 0.707s$.

Concept: Fillet welds fail in shear across the throat, the smallest section of the weld.

$P = 90 \times 0.707 \times 8 \times 150 = 76\,356$ N = **76.36 kN**.

Exam tip: Using the leg size s instead of the throat (108 kN) is the classic error. For a butt weld $P = \sigma\, t\, L$ with t = plate thickness.



Q10 **Answer: 26.16 (accept 26.03 to 26.29)**

Formula: Water (hydraulic) power = ρgQH; shaft power = ρgQH/η for a pump (and ηρgQH for a turbine).

Concept: ρgQH = 1000 × 9.81 × 0.05 × 40 = 19 620 W.

Shaft power = 19.62/0.75 = **26.16 kW**.

Exam tip: Divide by η for a pump (input is larger) and multiply for a turbine (output is smaller). Pump specific speed $N_s = N\sqrt{Q}/H^{3/4}$; turbine $N_s = N\sqrt{P}/H^{5/4}$.

Q11 **Answer: (A) 2, 4 and 8 times**

Formula: Affinity laws for one pump: $Q \propto N$, $H \propto N^2$, $P \propto N^3$ (for a changed impeller size at the same N: $Q \propto D^3$, $H \propto D^2$, $P \propto D^5$).

Concept: Similar velocity triangles: all velocities scale with N, so $Q \propto N$, $H \propto (\text{velocity})^2 \propto N^2$, $P \propto QH \propto N^3$.

Doubling N: Q × 2, H × 4, P × 8: **(A)**.

Exam tip: General scaling: $Q/(ND^3)$, $gH/(N^2D^2)$ and $P/(\rho N^3D^5)$ stay constant between geometrically similar machines.

Q12 **Answer: 44.36 (accept 44.14 to 44.58)**

Formula: $(T - T_\infty)/(T_i - T_\infty) = e^{-t/\tau}$, with time constant $\tau = \rho cV/(hA)$; valid when $Bi = hL_c/k < 0.1$, $L_c = V/A$.

Concept: After $t = 2\tau$ the excess temperature falls to $e^{-2} = 0.1353$ of its initial value.

$T = 20 + (200 - 20) \times 0.1353 = 20 + 24.36 =$ **44.36 °C**.

Exam tip: After one time constant 36.8 % of the initial excess remains (63.2 % of the change is complete). $L_c = r/3$ for a sphere, $r/2$ for a long cylinder, half-thickness for a plate.

Q13 **Answer: (D) 0.198 kJ/kg·K**

Formula: $\Delta s = c_p \ln(T_2/T_1) - R \ln(p_2/p_1) = c_v \ln(T_2/T_1) + R \ln(v_2/v_1)$.

Concept: Entropy is a property, so Δs depends only on the end states and not on the path.

$\Delta s = 1.005 \ln(5/3) - 0.287 \ln 3 = 0.5134 - 0.3153 =$ **0.198 kJ/kg·K (D)**.

Exam tip: A pressure rise lowers entropy, so the $R \ln(p_2/p_1)$ term is subtracted; adding it gives 0.828. For an incompressible solid or liquid, $\Delta s = c \ln(T_2/T_1)$.

Q14 **Answer: (A) 31°**

Formula: Merchant: $2\varphi + \beta - \alpha = 90°$, i.e. $\varphi = 45° + \alpha/2 - \beta/2$. Also $\tan\varphi = r\cos\alpha/(1 - r\sin\alpha)$, with chip thickness ratio $r = t/t_c$.

Concept: Merchant assumed that the shear plane sets itself where the cutting energy is minimum.

$\varphi = 45° + 4° - 18° =$ **31° (A)**.

Exam tip: A larger rake angle or a smaller friction angle gives a larger shear angle, a thinner chip and a lower cutting force. Shear strain $\gamma = \cot\varphi + \tan(\varphi - \alpha)$.

Q15 **Answer: 9**

Formula: For M/M/1 with $\rho = \lambda/\mu$: $L_s = \lambda/(\mu - \lambda)$, $L_q = \lambda^2/[\mu(\mu - \lambda)]$, $W_s = 1/(\mu - \lambda)$, $W_q = \lambda/[\mu(\mu - \lambda)]$; Little's law $L = \lambda W$.

Concept: $\rho = 15/20 = 0.75 < 1$, so the queue is stable.

$W_q = 15/(20 \times 5) = 0.15$ h = **9 minutes** ($W_s = 1/5$ h = 12 min, $L_q = 2.25$ trucks).

Exam tip: $W_s - W_q = 1/\mu$, the mean service time (3 min here). The probability that the bay is idle is $1 - \rho = 0.25$.

What next?

Day 89 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124



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